Variance-Reduced Manifold Sampling via Polynomial-Maximization Density Estimation

Serhii Zabolotnii*

Department of Information, Multimedia Technologies and Design,
Cherkasy State Business College, Cherkasy, 18028, Ukraine
State Scientific Research Institute of Armament and Military Equipment Testing and Certification,
Cherkasy, Ukraine
Department of Cybernetics and Applied Mathematics,
Uzhhorod National University, Uzhhorod, Ukraine

Preprint, 2026-05-19

Abstract

Uniform sampling on implicitly defined manifolds is a core primitive in motion planning, constrained simulation, and probabilistic machine learning. MASEM addresses this problem by entropy-maximizing resampling, but its resampling weights depend on a local k-nearest-neighbour density estimate whose errors can be amplified by aggressive resampling temperatures. We ask whether a polynomial-maximization moment estimator can replace the plug-in density rule without changing the surrounding MASEM architecture. The proposed PMM-MASEM module computes shell spacings from nested k-nearest-neighbour radii, estimates their standardized cumulants, and uses a gated PMM2/PMM3 estimator only when the spacing distribution departs from the flat Exp(1) regime; otherwise it falls back to the plug-in/MLE rule. This fallback is essential: on a flat homogeneous manifold the plug-in estimator is already the MLE, so PMM should not outperform it. A local Known-DGP Monte Carlo experiment confirms this gate: the selector returns MLE on flat Exp(1) spacings and reduces density MSE by 22–36% on asymmetric gamma and boundary-spacing regimes. The evidence is not uniformly positive: PMM3 worsens a platykurtic uniform spacing law, and a lightweight resampling-proxy experiment improves seven-lobes coverage but degrades the sine and swiss-roll proxies. The current evidence therefore supports an applicability-boundary result rather than a general MASEM improvement claim.

Keywords: density estimation; manifold sampling; resampling; k-nearest neighbours; simulation; reproducibility.

1 Introduction

Uniform sampling on constrained manifolds appears whenever a system is specified by equalities and inequalities rather than by an explicit parametrization. Typical examples include feasible robot configurations, implicit generative constraints, molecular or mechanical systems with conservation laws, and posterior slices in probabilistic models. The difficulty is not only geometric. A sampler

*ORCID: 0000-0003-0242-2234. Corresponding author: zabolotnii.serhii@csbc.edu.ua.

must represent all connected components of the feasible set and redistribute mass toward underrepresented regions without collapsing onto the components that happen to be easiest to reach from the initialization.

Manifold Sampling via Entropy Maximization (MASEM) [2] addresses this problem by alternating local manifold-constrained rejuvenation with an entropy-based resampling step. If x_i denotes a particle and $\varepsilon_{i,k}$ its distance to the k-th nearest neighbour, MASEM uses a weight of the form

$$w_i \propto \varepsilon_{i,k}^{\tau},$$

equivalently $w_i \propto \hat{q}(x_i)^{-\tau}$ for the plug-in density estimate

$$\hat{q}(x_i) = \frac{k}{N V_p \varepsilon_{i,k}^{p}}.$$

This rule gives larger weights to particles in low-density regions and is the mechanism by which MASEM spreads mass across disconnected components. It is also the statistical bottleneck. Braun et al. explicitly note that larger temperatures "can amplify errors in the k-NN density estimate" [2]; their practical mitigation is to ensemble several neighbourhood sizes, which they report as lower-variance but biased.

We ask whether that density-estimation subroutine can be replaced by a more principled moment-based estimator while preserving the rest of MASEM. The candidate is the polynomial maximization method (PMM), a family of closed form estimators for non-Gaussian errors developed in the Kunchenko school of moment estimation [6, 12]. In its PMM2 form, the standard coefficient

$$g = 1 - \frac{c_3^2}{2 + c_4}$$

relates estimator variance to the standardized third and fourth cumulants (c_3, c_4) of the error distribution. The MASEM density problem is not a direct regression problem, so this coefficient cannot be imported blindly. Instead, we use it as a diagnostic and design principle for a density module based on the shell spacings

$$\Delta_{i,j} = V_p \left(\varepsilon_{i,j}^{p} - \varepsilon_{i,j-1}^{p} \right).$$

A central constraint is the flat case. On a flat homogeneous manifold, the normalized shell spacings are asymptotically i.i.d. Exp(1) and the plug-in estimator is exactly the MLE. In that regime PMM has no statistical room to improve the density estimate. Our implementation therefore treats the flat Exp(1) cumulant signature $(c_3, c_4) = (2, 6)$ as an explicit fallback condition. The proposed PMM2/PMM3 rule is activated only when the observed spacing distribution shows curvature or boundary misspecification.

The contributions are:

1. a PMM-gated replacement for the MASEM density weight that preserves the NHR/OLLA rejuvenation architecture and changes only the resampling density estimate;

2. an explicit applicability boundary showing why flat homogeneous manifolds must fall back to the plug-in/MLE estimator;

3. a verification protocol and local evidence showing when the PMM gate helps, when it falls back, and when it fails;

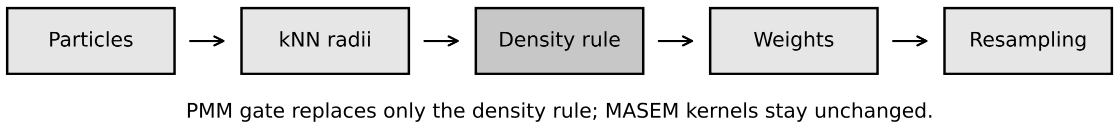

Figure 1: PMM-MASEM pipeline. The PMM gate replaces only the density rule used for resampling; the surrounding MASEM rejuvenation architecture is unchanged.

4. a reproducibility plan with fixed seeds, generated tables and figures, and an AI-use disclosure for the final manuscript bundle.

The resulting evidence is deliberately mixed: the gate preserves the flat MLE case and improves local density MSE in asymmetric spacing regimes, while proxy experiments expose failure modes that preclude a general MASEM-superiority claim.

Before describing the method, we position the work relative to manifold sampling, nonparametric density estimation, and PMM moment estimators (§2).

2 Related Work

Manifold sampling and entropy resampling. Sampling on implicitly defined manifolds has been approached through rejection methods, constrained Markov chains, sequential Monte Carlo, and local tangent-space kernels. MASEM [2] is the direct starting point for this paper because it addresses disconnected feasible sets by combining local feasible rejuvenation with entropy-driven resampling. Its resampling step uses an empirical density surrogate to move probability mass towards underrepresented regions. The present work does not replace the MASEM architecture, local kernels, or constraint handling: it isolates the statistical role of the density estimate inside the resampling weights.

k-nearest-neighbour density estimation. The MASEM plug-in rule descends from classical multivariate k-nearest neighbour density estimation [7]. Mack and Rosenblatt [8] analysed its bias–variance behaviour and emphasized the dependence on k, tail behaviour, and the local approximation of density by a ball volume. Recent analyses sharpen this picture by separating risk criteria and support assumptions: for example, Zhao and Lai [15] show that boundary knowledge can change whether kNN density estimation is minimax-optimal under strong uniform criteria. This literature sets the baseline for our method: a single-radius plug-in estimator is strong in the locally Euclidean case, whereas boundary or support misspecification can create a real opportunity for a guarded replacement.

Neighbour distances, spacings, and the Poisson null model. Nearest-neighbour distances also underlie spacing and entropy estimators. The Kozachenko–Leonenko entropy estimator [5] uses kNN radii to estimate local log-density, while the classical theory of spacings [11] and Poisson processes [4, 10] gives the reference law for shell-volume increments. In a locally homogeneous flat model, the increments of nested ball volumes behave as independent exponential spacings after the local intensity is factored out. This Poisson case is the null model of the present paper: under Exp(1) spacings the plug-in density rule is also the MLE, so any PMM module must fall back rather than claim improvement.

Density estimation on manifolds and near boundaries. Non-Euclidean density estimation makes the preceding local-ball approximation more delicate. Pelletier [9] adapts kernel density estimation to compact Riemannian manifolds using geodesic structure and volume corrections; Henry, Muñoz, and Rodriguez [3] study kNN-type density estimation when observations belong to a Riemannian manifold. Boundary effects form a separate source of misspecification: Berry and Sauer [1] develop density estimators for manifolds with boundary by estimating boundary direction and distance. These works motivate our experimental regimes. Curvature and boundary truncation are not incidental nuisances for MASEM; they are precisely the mechanisms that can move shell-spacing cumulants away from the flat exponential reference law.

Moment estimators and PMM. The polynomial maximization method (PMM) provides moment-based corrections for non-Gaussian error regimes through standardized cumulants [6, 12]. The PMM2 coefficient $g = 1 - c_3^2/(2 + c_4)$ is attractive here because the shell-spacing distribution can be summarized by (c_3, c_4) before a density rule is chosen. Recent preprints and software work extend this estimator family to non-Gaussian regression and time-series settings, including the EstemPMM implementation and a large PMM2–ARIMA simulation study [13, 14]. We use these works as documentation that PMM2/PMM3 form an implementable cumulant-based estimation family, not as direct evidence for the MASEM density task. That task is not a standard linear regression, measurement, or time-series problem. The present paper is therefore an applicability study, not a direct transplantation of PMM theory: when the spacing law is Exp(1) the plug-in/MLE rule is retained; when curvature or boundary effects move the cumulants away from that regime, PMM2/PMM3 become guarded candidates. Adaptive fractional bases and PATP-style selectors are left to a separate follow-up study rather than included in the present evidence package.

These observations motivate a theoretical analysis of k-NN shell spacings on flat and curved manifolds (§3).

3 Theoretical Framework

This section isolates the statistical object modified by PMM-MASEM: the local density estimate inside MASEM's resampling step. The surrounding manifold kernel is not changed.

3.1 Flat Homogeneous Manifold

Let $\mathcal{M}$ be a p-dimensional flat manifold and let ρ be locally constant at the scale of the k-NN ball around x_i. Define shell spacings

$$\Delta_{i,j} = V_p(\varepsilon_{i,j}^p - \varepsilon_{i,j-1}^p), \qquad \varepsilon_{i,0} = 0.$$

Under the standard Poisson approximation for local neighbour counts, the normalized spacings

$$s_{i,j} = N\rho(x_i)\Delta_{i,j}$$

are asymptotically i.i.d. Exp(1) as $N \to \infty$ with fixed k. Consequently, $\mathbb{E}[s_{i,j}] = 1$, $\mathrm{Var}(s_{i,j}) = 1$, and the standardized cumulants are $c_3 = 2$, $c_4 = 6$.

Proposition 1 (Plug-in estimator is the flat-case MLE)**.** *Under the flat homogeneous assumptions, the MLE for $\rho(x_i)$ based on the shell spacings is*

$$\hat{\rho}_{\mathrm{MLE}}(x_i) = \frac{k}{N\sum_{j=1}^{k}\Delta_{i,j}} = \frac{k}{NV_p\varepsilon_{i,k}^p},$$

which is exactly the MASEM plug-in estimator.

The first consequence is methodological: a PMM replacement should not be applied in this regime. Any claimed improvement over the plug-in estimator on a truly flat homogeneous manifold would be inconsistent with the likelihood solution under the correct model. Our algorithm therefore uses the empirical cumulant signature $(c_3, c_4) \approx (2, 6)$ as a fallback gate.

3.2 Curvature and Boundary Misspecification

On a Riemannian manifold the Euclidean volume term $V_p r^p$ is only a local approximation to the geodesic ball volume. For a constant-curvature patch,

$$\mathrm{vol}_g(B(x,r)) = V_p r^p \left(1 - \frac{\kappa r^2}{6(p+2)} + O(\kappa^2 r^4)\right),$$

with boundary terms adding further non-Euclidean corrections. The plug-in estimator still uses the flat $V_p r^p$ volume, so it becomes an MLE for a misspecified model. The resulting spacing distribution need not be Exp(1); its cumulants shift away from $(2, 6)$ in a manner governed by curvature, boundary truncation, density variation, k, and N.

Accordingly, the main text uses a conservative qualitative claim: PMM is admissible only when the measured shell-spacing law departs from the flat signature. The exact leading-order formulas in `theory/riemannian_manifold.tex` require a separate audit before they should be promoted as theorem-level claims. The experiments in §5 are designed to validate or falsify those formulas on known-DGP patches.

3.3 From Density Error to Component Loss

MASEM uses weights $w_i \propto \hat{\rho}(x_i)^{-\tau}$. If $\hat{\rho}(x_i) = \rho(x_i)(1 + \varepsilon_i)$, then

$$\hat{\rho}(x_i)^{-\tau} \approx \rho(x_i)^{-\tau}\left(1 - \tau\varepsilon_i + \frac{\tau(\tau+1)}{2}\varepsilon_i^2\right).$$

Thus density-estimation variance is magnified by τ^2 in the weight variance. This is the technical reason PMM, if it reduces density error in a misspecified curved regime, could allow a larger safe temperature.

For a component $\mathcal{C}_c$ with effective mass floor $(\Phi\alpha)_c$, the usual occupancy argument gives a component-loss bound of the form

$$P_c \le \exp\{-N(\Phi\alpha)_c\}.$$

The role of the density estimator is indirect: lower weight noise increases the range of τ for which the effective floor remains reliable. The final paper reports this as a conditional bridge, not as a new convergence theorem, unless the proof is fully completed.

This analysis leads to the PMM module and its integration into MASEM.

Table 1: Empirical kNN-spacing regimes in the Known-DGP Monte Carlo. Values are means over five seeds; $g_2 = 1 - c_3^2/(2 + c_4)$ is diagnostic and is not interpreted as an end-to-end gain.

Regime	c_3	c_4	g_2	selector
Flat Exp(1)	1.962	5.624	0.495	MLE_fallback
Mild curved gamma	2.410	8.841	0.460	PMM2
Strong curved gamma	3.010	12.981	0.394	PMM2
Boundary mixture	2.272	8.442	0.494	PMM2
Platykurtic uniform	0.009	-1.191	1.000	PMM3

4 Method

PMM-MASEM changes one part of MASEM: the density estimate used to compute resampling weights. Initialization, local manifold rejuvenation, NHR/OLLA kernels, and slack handling remain unchanged.

4.1 Baseline MASEM Weight

For particles $x_1, \dots, x_N$ on a p-dimensional manifold, baseline MASEM computes the k-NN radius $\varepsilon_{i,k}$ and uses

$$\hat{\rho}_{\text{Plugin}}(x_i) = \frac{k}{N V_p \varepsilon_{i,k}^p}, \qquad w_i = \frac{\hat{\rho}_{\text{Plugin}}(x_i)^{-\tau}}{\sum_{\ell=1}^N \hat{\rho}_{\text{Plugin}}(x_\ell)^{-\tau}}.$$

The PMM module replaces only $\hat{\rho}_{\text{Plugin}}$.

4.2 Shell Spacings and Cumulants

For each particle we compute sorted radii $\varepsilon_{i,1} \le \cdots \le \varepsilon_{i,k}$ and shell spacings

$$\Delta_{i,j} = V_p(\varepsilon_{i,j}^p - \varepsilon_{i,j-1}^p).$$

The normalized spacings are pooled across particles to estimate standardized cumulants (c_3, c_4). Pooling is necessary because a single particle gives only k spacings, which are too few for stable cumulant estimation. The current implementation uses global pooling; component-aware or local pooling is an experimental extension.

4.3 Selector

The selector has three gates.

Flat gate. If (c_3, c_4) is close to $(2, 6)$, the module returns the MLE/plug-in density. This gate enforces the flat-manifold applicability boundary.

PMM2 gate. If the spacing law is not Exp(1)-like and the skewness gate is active, the module computes

$$\hat{\rho}_{\text{PMM2}}(x_i) = \frac{1}{N \bar{\Delta}_i} \left(1 + \frac{c_3(m_{1i} - m_1)}{2 + c_4} \right),$$

where $\bar{\Delta}_i = k^{-1} \sum_j \Delta_{i,j}$, m_{1i} is the per-particle mean of normalized spacings, and m_1 is the pooled mean.

PMM3 gate. If the skewness is small but the spacing law is platykurtic, the module uses a PMM3 correction based on the per-particle second moment. Both PMM gates are guarded by denominator and variance-coefficient checks. Invalid regimes fall back to MLE.

4.4 Algorithm

Algorithm 1: PMM density module for MASEM

Input: particles $x_{1:N}$, intrinsic dimension p, neighbourhood size k, temperature τ.
Output: normalized resampling weights $w_{1:N}$.

1. For each particle x_i, compute sorted radii $\varepsilon_{i,1} \leq \cdots \leq \varepsilon_{i,k}$.
2. Convert radii to shell spacings $\Delta_{i,j} = V_p(\varepsilon_{i,j}^p - \varepsilon_{i,j-1}^p)$.
3. Normalize spacings $s_{i,j} = N\Delta_{i,j}$ and pool them over all i, j.
4. Estimate pooled standardized cumulants (c_3, c_4) from $s_{i,j}$.
5. Compute the fallback density $\hat{\rho}_{\mathrm{MLE},i} = 1/(N\bar{\Delta}_i)$.
6. **if** (c_3, c_4) is Exp(1)-like, i.e. close to $(2, 6)$, **then**
set $\hat{\rho}_i \leftarrow \hat{\rho}_{\mathrm{MLE},i}$.
7. **else if** the PMM2 denominator and variance checks are valid **then**
set $\hat{\rho}_i \leftarrow \hat{\rho}_{\mathrm{PMM2},i}$.
8. **else if** $|c_3|$ is small, $c_4 < 0$, and the PMM3 checks are valid **then**
set $\hat{\rho}_i \leftarrow \hat{\rho}_{\mathrm{PMM3},i}$.
9. **else**
set $\hat{\rho}_i \leftarrow \hat{\rho}_{\mathrm{MLE},i}$.
10. Return $w_i = \hat{\rho}_i^{-\tau} / \sum_{\ell=1}^{N} \hat{\rho}_\ell^{-\tau}$.

4.5 Implementation Status

The repository currently implements the selector in `src/masem/pmm_module.py`, the estimator registry in `src/masem/estimators.py`, and MASEM integration in `src/masem/masem.py`. Unit tests validate shape, normalization, JAX tracing, fallback behaviour, and PMM2/PMM3 activation on synthetic spacing laws.

§5 verifies the theoretical predictions on synthetic and benchmark manifolds.

5 Experiments

The experiments are designed to answer two questions separately. First, does the PMM selector improve local density estimation when the spacing law is known to deviate from Exp(1)? Second, does any local improvement translate into end-to-end MASEM sampling quality?

5.1 Experimental Setup

The four estimators are fixed throughout:

$$Plugin\_Estimator, \quad k\_Ensemble, \quad MLE - Exp, \quad PMM2/PMM3.$$

We evaluate two empirical layers. The first layer is a Known-DGP density Monte Carlo in `experiments/run_known_dgp_mc.py`. It generates flat Exp(1), asymmetric gamma, boundary-mixture, and platykurtic spacing laws with known target density. Only this layer is used to support a local density-estimation claim.

The second layer is a lightweight resampling proxy in `experiments/run_resampling_proxy.py`. It uses biased particle clouds on disconnected disks, seven lobes, a sine curve, a swiss roll, a scaling stress test, and a robotics-corridor proxy, then applies the four density weight rules to a matched resampling step. This layer tests whether the weight rule has a plausible operational effect, but it is not a full MASEM NHR/OLLA benchmark. We therefore treat it as operational diagnostic evidence rather than as an end-to-end superiority benchmark.

Each experiment uses five seeds and reports mean ± 95% confidence intervals. Pairwise significance tests use Holm–Bonferroni correction with threshold $p < 0.01$. Metrics are:

- Wasserstein-2 squared distance (Sinkhorn approximation), the primary end-to-end metric;
- KL divergence between pairwise-distance histograms;
- maximum slack and feasibility diagnostics;
- component-loss frequency;
- wall-clock overhead.

5.2 Known-DGP Density Microbenchmark

Known-DGP patches isolate density estimation from MASEM dynamics. The flat patch validates the fallback rule: Exp(1)-like spacings should select Plugin/MLE. Curved and boundary patches test whether PMM2/PMM3 reduce density MSE relative to Plugin and k-ensemble baselines. The generated output is `results/known_dgp_mc.csv`.

5.3 Resampling-Proxy Sampling

The proxy experiments compare the four density rules under matched particle budgets, k, τ, and synthetic target distributions. They report nearest-neighbour W_2^2 proxy, pairwise-distance histogram KL, component loss, and estimator wall-clock. Since no local manifold rejuvenation is run, the slack metric is fixed at zero and the results are interpreted only as weight-rule diagnostics.

5.4 τ-Tolerance

The τ sweep tests whether lower density-estimation error permits a larger temperature before W2 or component loss degrades. This is the operational meaning of the theory bridge in §3; the output is `results/tau_tolerance.csv`.

5.5 Component Coverage

Component coverage is evaluated by tracking whether each disconnected component or lobe receives at least one final particle. For synthetic tasks with known components, this is direct. For swiss roll and robotics tasks, the component/lobe proxy is specified before running the experiment.

5.6 Computational Overhead

The PMM module adds cumulant estimation and correction terms to the density step. Since all operations are vectorized, the expected overhead is moderate, but the wall-clock effect is measured empirically. Proxy wall-clock results are stored in `results/wallclock.csv`.

All CSV files, tables, and figures are generated by the repository-wide experiment driver. §6 summarizes the resulting evidence and states the remaining end-to-end gap explicitly.

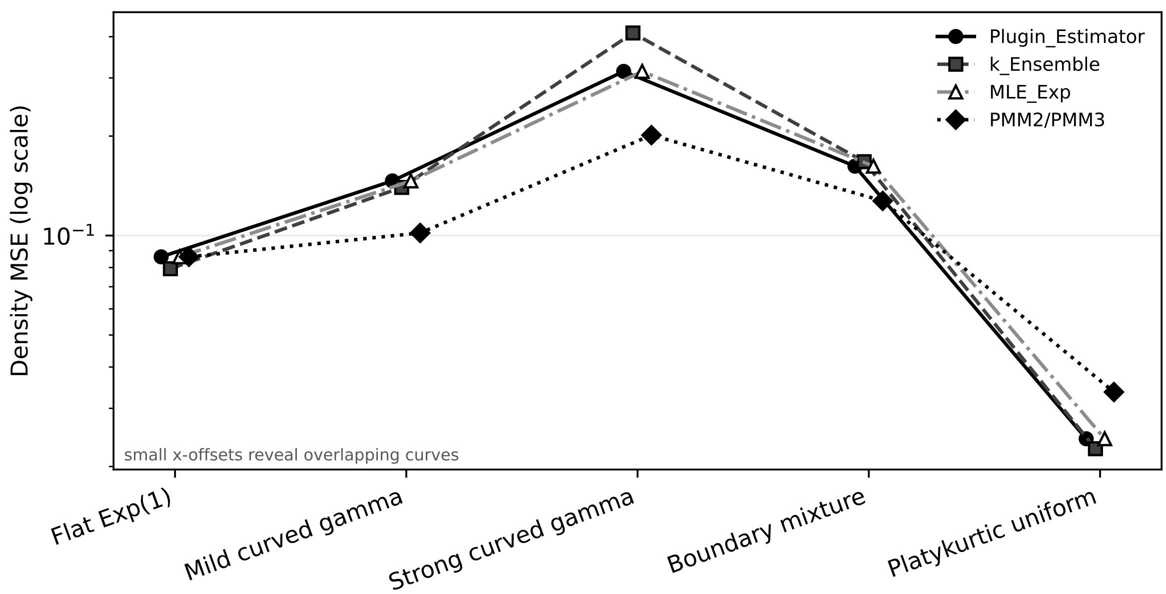

Figure 2: Known-DGP density MSE by spacing regime. PMM2 reduces MSE on asymmetric gamma and boundary regimes while preserving the flat MLE fallback; PMM3 degrades the platykurtic uniform stress case. Small horizontal offsets are used only to reveal estimators whose values overlap exactly.

6 Results

H1: local density estimation. The Known-DGP microbenchmark supports the intended flat fallback and gives a conditional positive result for asymmetric misspecification. On flat Exp(1) spacings, the selector chooses MLE in all five seeds and the PMM2/PMM3 MSE ratio against MLE is exactly 1.000. On asymmetric regimes the selected PMM rule reduces density MSE: the ratio is 0.688 on mild gamma spacings, 0.639 on strong gamma spacings, and 0.785 on the boundary-mixture proxy. The platykurtic uniform stress case is negative: PMM3 is selected, but its MSE ratio is 1.385. Thus H1 is partially confirmed for PMM2-style asymmetric regimes and not supported for the current PMM3 branch.

H2: resampling quality. The proxy layer does not establish end-to-end MASEM superiority. It does show that the density rule can matter operationally. PMM improves the seven-lobes proxy from 0.0074 to 0.0044 mean W_2^2 with Holm-adjusted $p = 0.001$, but it degrades the sine and swiss-roll proxies. The swiss-roll degradation is clearest: PMM increases proxy W_2^2 from 0.158 to 0.711 with adjusted $p = 0.001$. These results support an applicability boundary rather than a positive end-to-end result.

τ **tolerance.** The proxy τ sweep is consistent with the mixed result. On seven lobes, PMM remains within 10% of its $\tau = 0.5$ value up to the largest tested $\tau = 1.5$, whereas Plugin and MLE tolerate only $\tau = 0.5$ under the same criterion. On swiss roll and the robotics proxy, all estimators tolerate the tested range, so this diagnostic does not distinguish them.

Table 2: Single-step resampling-proxy W_2^2 results. Entries are mean $\pm$ 95% CI over five seeds; parentheses report Holm-adjusted paired p-values against Plugin. This table is weight-rule evidence, not a full NHR/OLLA MASEM benchmark.

Benchmark	Plugin	k-Ens.	MLE-Exp	PMM2/PMM3
Disconnected disks	0.004 ± 0.000	0.004 ± 0.000 (0.886)	0.004 ± 0.000 (–)	0.003 ± 0.000 (0.032)
Seven lobes	0.007 ± 0.002	0.005 ± 0.001 (0.004)	0.007 ± 0.002 (–)	0.004 ± 0.001 (0.001)
Sine curve	4.905 ± 2.884	4.905 ± 2.884 (0.392)	4.905 ± 2.884 (–)	37.829 ± 18.495 (0.020)
Swiss roll	0.158 ± 0.026	0.158 ± 0.026 (0.952)	0.158 ± 0.026 (–)	0.711 ± 0.141 (0.001)
Scaling stress-test	0.013 ± 0.005	0.013 ± 0.005 (0.198)	0.013 ± 0.005 (–)	0.013 ± 0.005 (0.672)
Robotics corridor proxy	0.003 ± 0.001	0.003 ± 0.001 (0.126)	0.003 ± 0.001 (–)	0.003 ± 0.001 (0.020)

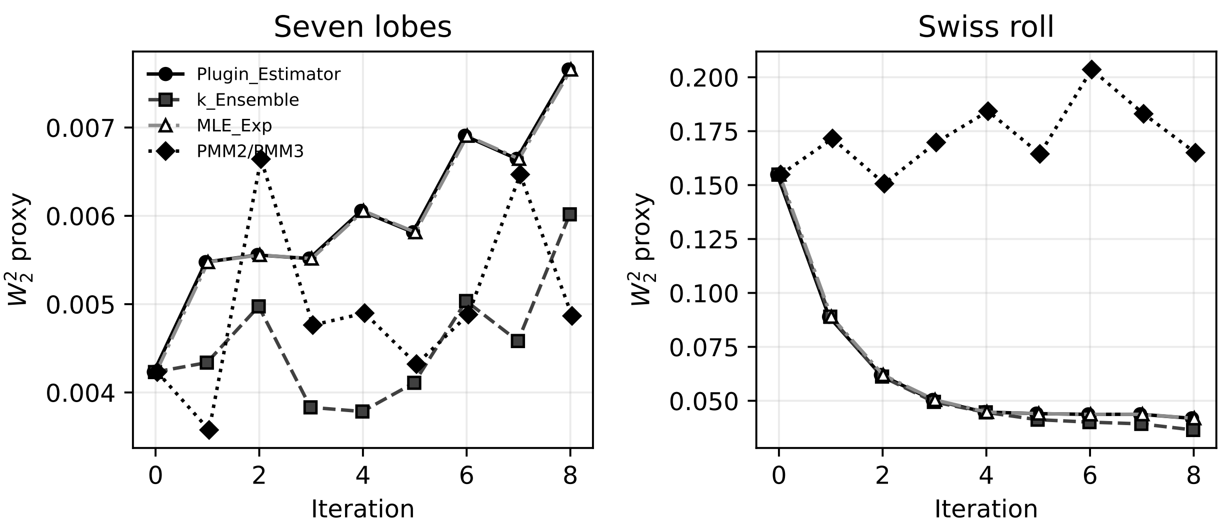

Figure 3: Iterated resampling proxy for seven lobes and swiss roll. The matched-refresh proxy is diagnostic only; it does not replace a full MASEM NHR/OLLA benchmark. Small horizontal offsets reveal overlapping Plugin/MLE curves.

H3: component loss. Component coverage does not support a general PMM advantage. PMM has no loss at $N = 220$ in the component sweep, but at lower budgets it loses components more often than the other estimators. In the main proxy table, sine and swiss-roll coverage also worsens under PMM. H3 is therefore not supported.

Computational overhead. In these small unjitted proxy runs, PMM is not the wall-clock bottleneck: its mean estimator time is about 0.40 times the Plugin measurement, largely because the Plugin path pays for the full k-NN radius computation while the PMM/MLE paths reuse shell-spacing structure efficiently. This estimate should not be generalized to a compiled full MASEM run, but it indicates that the PMM gate is not currently a dominant cost.

H4: GSA schedule. The adaptive GSA schedule was not implemented in the present evidence bundle. H4 is therefore not evaluated and remains future work.

Overall, RQ1 is partially confirmed: PMM2 can reduce local density MSE under asymmetric spacing misspecification while correctly falling back on flat Exp(1). RQ2 is not supported as an

Table 3: Proxy τ tolerance: largest tested τ whose mean W_2^2 stays within 10% of the estimator's $\tau = 0.50$ value.

Benchmark	Plugin	k-Ens.	MLE-Exp	PMM2/PMM3
Seven lobes	0.50	0.75	0.50	1.50
Swiss roll	1.50	1.50	1.50	1.50
Robotics corridor proxy	1.50	1.50	1.50	1.50

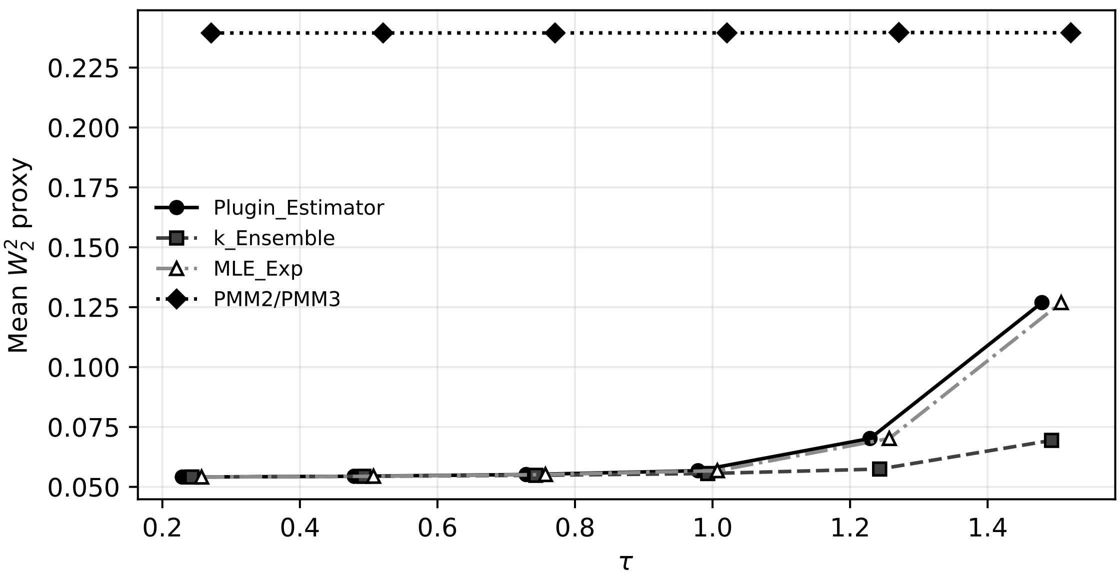

Figure 4: Proxy W_2^2 as a function of resampling temperature τ. Small horizontal offsets reveal overlapping Plugin/MLE curves.

end-to-end MASEM claim: the current proxy evidence is mixed and exposes failure modes on sine, swiss-roll, and PMM3 platykurtic regimes.

7 Discussion

The present evidence validates PMM-MASEM as a local diagnostic and density-estimation module rather than as a complete sampler improvement. Its main scientific value is a sharper statistical account of when the MASEM density estimate has room for improvement. In the flat homogeneous case, the boundary is clear: the plug-in estimator is the MLE and should be retained. Any useful PMM effect must come from curvature, boundary truncation, component imbalance, or other misspecification of the Exp(1) spacing model.

This framing determines how negative results should be read. If PMM improves the known-DGP density task but not end-to-end sampling, then the local density effect is real but not operationally decisive under the tested MASEM dynamics. If PMM does not improve even the known-DGP task, then the PMM moment correction is not appropriate for this spacing model as currently formulated. If PMM helps only under artificial curvature or boundary strength, then the result is an applicability map rather than a general-purpose sampler improvement.

The evidence narrows the claim while strengthening its interpretation. The Known-DGP exper-

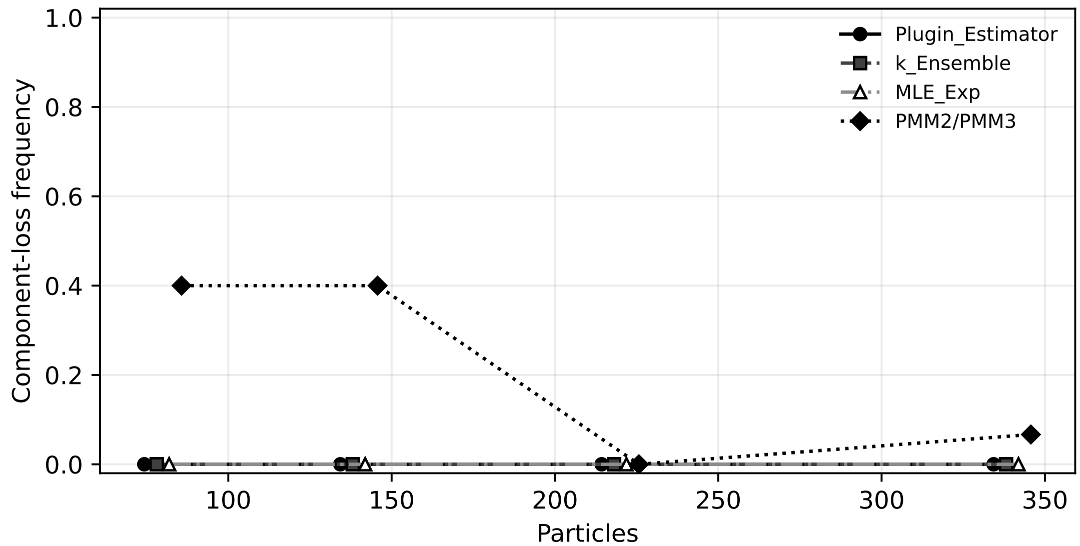

Figure 5: Component-loss frequency in the proxy sweep. Lower values are better. Small horizontal offsets reveal overlapping zero-loss curves.

iment justifies PMM2 as a variance-reduction candidate under asymmetric spacing misspecification, yet the PMM3 stress case and the sine/swiss-roll proxy failures show that the selector is insufficient to support a broad "PMM improves MASEM" conclusion. This should be treated as the main scientific message: the kNN density bottleneck is real, but moment corrections must be regime-aware and can be harmful when the spacing law violates the PMM branch assumptions.

The most important implementation limitation is cumulant estimation. Global pooling is stable but may blur component-specific geometry. Local pooling is more faithful but has higher variance. Component-aware pooling after a coarse partition may offer a useful compromise, but it would add another algorithmic choice that must be validated. The PMM3 branch is also intentionally conservative; it should remain a fallback branch unless a real benchmark exhibits a symmetric platykurtic spacing law.

The theoretical limitation is the status of the curved-manifold cumulant formulas. The flat Exp(1) result is standard and strong. The curvature corrections are currently best treated as heuristic unless the derivation is audited and matched by known-DGP simulations. We therefore avoid theorem-level claims based on these formulas.

Future work should first replace the proxy layer with the full MASEM NHR/OLLA benchmark suite, then explore component-aware cumulant pooling, adaptive temperature schedules, and alternative moment estimators for boundary-heavy manifolds. GSA-based stopping remains a secondary axis: it belongs in the appendix unless it demonstrates a clear advantage in reproducible experiments.

§8 summarizes the contribution.

8 Conclusion

This paper develops a PMM-gated density module for MASEM. The module replaces only the resampling density estimate, computes shell-spacing cumulants, falls back to Plugin/MLE in the flat Exp(1) regime, and activates PMM2/PMM3 only under measured misspecification. The central

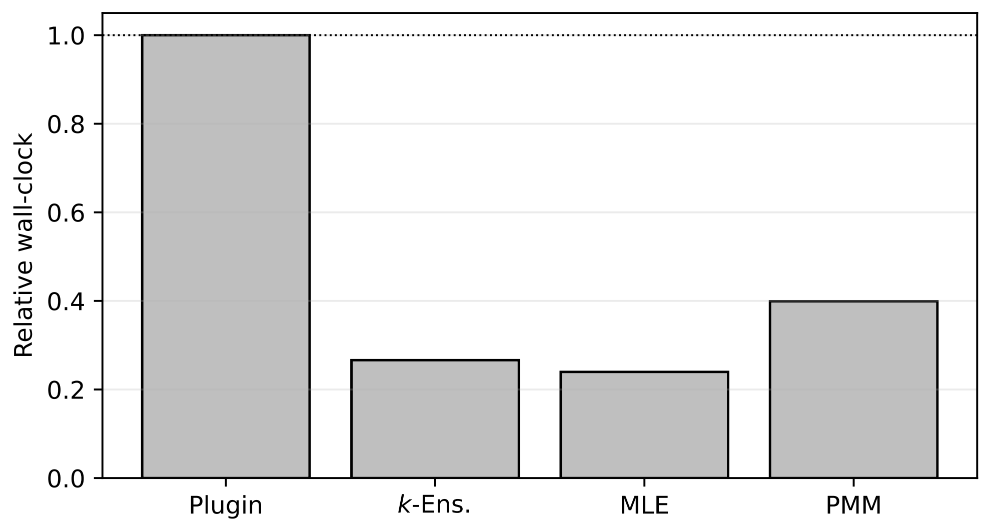

Figure 6: Relative estimator wall-clock in the proxy experiment.

boundary is explicit: PMM should not be expected to outperform the plug-in estimator on a flat homogeneous manifold because the plug-in rule is already the MLE.

The current experiments support a boundary-paper conclusion. PMM2 reduces local density MSE on asymmetric gamma and boundary-spacing DGPs, but PMM3 degrades performance in a platykurtic stress case and the resampling proxy is mixed: the seven-lobes proxy improves, while sine and swiss roll degrade. The contribution is therefore not a finished general-purpose sampler improvement. It is a validated diagnostic and replacement module that identifies where PMM has statistical room to help, where it must fall back, and where the present selector remains unreliable.

A Appendix Overview

This appendix records what must be included in the submission-ready version. Detailed derivation drafts currently live in `theory/flat_manifold.tex`, `theory/riemannian_manifold.tex`, and `theory/quantitative_bridge.tex`. They should be merged only after the g/variance wording is audited.

B A.1 Flat-Manifold Spacing Derivation

For a locally homogeneous flat manifold, the count process in balls around x_i converges to a Poisson process in the volume coordinate $u = N\rho(x_i)V_p r^p$. Consecutive arrivals in that coordinate have i.i.d. Exp(1) spacings. Since

$$N\rho(x_i)\Delta_{i,j} = N\rho(x_i)V_p(\varepsilon_{i,j}^p - \varepsilon_{i,j-1}^p),$$

the shell spacings inherit the Exp(1) law. Summing spacings gives $\sum_j \Delta_{i,j} = V_p\varepsilon_{i,k}^p$, hence the MLE is the plug-in density estimator.

C A.2 Curved-Manifold Heuristic

The curved-manifold analysis starts from the geodesic ball expansion

$$\mathrm{vol}_g(B(x,r)) = V_p r^p \left(1 - \frac{\kappa r^2}{6(p+2)} + O(\kappa^2 r^4)\right).$$

This changes the spacing law observed by a density estimator that still uses the Euclidean volume coordinate. The final version must decide whether to state the resulting cumulant shifts as theorem, approximation, or empirical diagnostic. Until Known-DGP MC confirms the formulas, this appendix should use conservative language.

D A.3 PMM Implementation Details

The implementation is vectorized in JAX. Pairwise distances are computed for all particles, sorted to obtain k-NN radii, and converted to shell spacings. The selector computes all candidate densities and uses `jnp.where` for traceable selection. The flat gate tests proximity to $(c_3, c_4) = (2, 6)$. Invalid PMM denominators, negative variance coefficients, and boundary cumulant regimes fall back to MLE.

E A.4 Full Tables

The generated main tables are included in the body as Tab. 1, Tab. 2, and Tab. 3. Their source CSVs are stored under `results/`: `known_dgp_mc.csv`, `main_benchmarks.csv`, and `tau_tolerance.csv`.

F A.5 GSA Adaptive Schedule

GSA is a secondary axis. Unless implemented and validated, the final appendix should state that adaptive stopping was not evaluated in the main empirical claim.

G A.6 Known-DGP Monte Carlo Protocol

Known-DGP experiments must include flat, curved, and boundary patches. The required outputs are density bias, density variance, MSE ratio relative to Plugin/MLE, estimated (c_3, c_4), and selector branch counts. Flat patches must trigger Plugin/MLE fallback.

H A.7 Reproducibility Manifest

The current repository includes the local driver `python3-mexperiments.run_all`, fixed seeds, generated result CSVs, tables, and figures. A public archive/DOI is still required before submission if mandated by the venue.

I A.8 AI-Use Disclosure

Drafting, code review, and planning assistance used an AI coding assistant. All mathematical claims, experimental results, and final wording must be reviewed and approved by the authors. No result should be reported unless it is generated by the repository scripts and traceable to saved artifacts.

References

[1] Tyrus Berry and Timothy Sauer. Density estimation on manifolds with boundary. *Computational Statistics & Data Analysis*, 107:1–17, 2017. doi: 10.1016/j.csda.2016.09.011.

[2] C. V. Braun, T. Burghoff, and M. Toussaint. Manifold sampling via entropy maximization. *arXiv preprint arXiv:2605.12338*, 2026.

[3] Guillermo Henry, Andrés Muñoz, and Daniela Rodriguez. k-nearest neighbor density estimation on riemannian manifolds. *arXiv preprint arXiv:1106.4763*, 2011.

[4] J. F. C. Kingman. *Poisson Processes*. Oxford University Press, 1993.

[5] L. F. Kozachenko and N. N. Leonenko. Sample estimate of the entropy of a random vector. *Problems of Information Transmission*, 23(2):95–101, 1987.

[6] Yu. P. Kunchenko. *Stochastic Polynomials*. Naukova Dumka, Kyiv, 2006.

[7] Don O. Loftsgaarden and Charles P. Quesenberry. A nonparametric estimate of a multivariate density function. *The Annals of Mathematical Statistics*, 36(3):1049–1051, 1965. doi: 10.1214/aoms/1177700079.

[8] Y. P. Mack and M. Rosenblatt. Multivariate k-nearest neighbor density estimates. *Journal of Multivariate Analysis*, 9(1):1–15, 1979. doi: 10.1016/0047-259X(79)90065-4.

[9] Bruno Pelletier. Kernel density estimation on riemannian manifolds. *Statistics & Probability Letters*, 73(3):297–304, 2005. doi: 10.1016/j.spl.2005.04.004.

[10] Mathew Penrose. *Random Geometric Graphs*. Oxford University Press, 2003.

[11] Ronald Pyke. Spacings. *Journal of the Royal Statistical Society. Series B (Methodological)*, 27(3):395–436, 1965. doi: 10.1111/j.2517-6161.1965.tb00602.x.

[12] Zygmunt Lech Warsza and Serhii V. Zabolotnii. Estimation of measurand parameters for data from asymmetric distributions by polynomial maximization method. In *Automation 2018*, volume 743 of *Advances in Intelligent Systems and Computing*, pages 746–757. Springer, Cham, 2018. doi: 10.1007/978-3-319-77179-3_74.

[13] Serhii Zabolotnii. Applying the polynomial maximization method to estimate ARIMA models with asymmetric non-Gaussian innovations. *arXiv preprint arXiv:2511.07059*, 2025. doi: 10.48550/arXiv.2511.07059. URL `https://arxiv.org/abs/2511.07059`.

[14] Serhii Zabolotnii. EstemPMM: Polynomial maximization method for non-gaussian regression and time series in R. *arXiv preprint arXiv:2605.02673*, 2026. doi: 10.48550/arXiv.2605.02673. URL `https://arxiv.org/abs/2605.02673`.

[15] Puning Zhao and Lifeng Lai. Analysis of KNN density estimation. *arXiv preprint arXiv:2010.00438*, 2020.